# Chapter 5: Nearest Neighbors and Ensembles

## Multiple choice questions

**Question 1**

Level: easy

Ensemble learners combine classifiers to increase the performance. Applying the same algorithm independently to each of several bootstrapped samples from the dataset with the same set of input variables and then aggregating the results by means of majority voting is called:

a) Random Forests
b) Boosting
c) Bagging.
d) Stacking.

**Question 2**

Level: easy

Which kind of models are typically improved by developing an ensemble of them:

a) Models with high bias
b) Models with high variance
c) Stable models
d) Models with high bias and variance

## Open questions

**Question 1**

Explain the bias-variance trade-off, and relate to the AIC and BIC measures.

**Question 2**

How are bagging, boosting and random forests different?

**Question 3**

Why do ensemble methods typically use decision trees, decision stumps or neural networks as base-learners?

**Question 4**

What is a meta-learner?

**Question 5**

What is the conceptual similarity between kNN and ensembles, and between KNN and a decision tree?